Reviewer Type:	**Indian**
Reviewer Name:	**Prof. Pinaki Majumdar**
Reviewer Desination:	**Professor**
Reviewer Email:	**pinaki.majumdar@ahduni.edu.in**
Reviewer Phone No.:	**079-61911516**
Reviewer Address:	**School of Arts and Sciences, Ahmedabad University Navrangpura, Ahmedabad 380009 Gujarat, India Phone: 079-61911516**
Reviewer Accepted/Rejected:	**Accepted with minor changes**
Reviewer Selected For Viva voce:	**No**

Assessment of thesis submitted by Abhirup Mukherjee

Abhirup Mukherjee has submitted a thesis called "A New Auxiliary Model Approach To Fermionic Criticality: Insights on Mottness in Strongly Correlated Systems". It is a long thesis, running to about 330 pages. In response, this is a long assessment.

This thesis presents an ambitious programme aimed at developing a Hamiltonian based renormalisation group (RG) framework for strongly correlated electron systems. The principal applications are to the Mott metal-insulator transition (MIT), pseudogap phenomena, heavy-fermion physics and, more briefly, holographic approaches to quantum matter. Through most of the thesis, the author seeks to construct a unified theoretical framework based on extended auxiliary impurity Hamiltonians, their analysis using the unitary renormalisation group (URG), and the subsequent reconstruction of lattice properties through a translation-operator formalism.

The problems addressed are central to correlated electron phenomena. In particular, the Hubbard model and its variants continue to be studied using a wide variety of analytical and numerical techniques, including dynamical mean field theory (DMFT), its cluster extensions, quantum Monte Carlo, tensor-network methods and several other high-accuracy approaches. Any new formulation of these benchmark problems carries a high burden of scientific justification.

The originality of the work lies in (i) the construction of extended auxiliary impurity Hamiltonians, (ii) the development of a systematic URG treatment of these models, and (iii) the proposal that their low-energy fixed-point structure provides an effective description of correlated lattice models. The technical competence of the work is high.

I would state that just for the combined results of (i) and (ii) above the thesis deserves the award of a PhD degree. With that out of the way we can now discuss the physics and point out where clarifications can help, particularly with respect to (iii).

Before getting into a chapter-wise discussion let me put up what I see as the strategy of the thesis, and then use the two dimensional Hubbard model as a test case to pose some questions. It is possible that the answers are embedded at various places in the lattice, but a restatement may nevertheless help people who trained in the mainstraem of correlated electron theory.

In my understanding, when the author aims to solve a lattice model the following steps are executed.

1. An 'extended' impurity model, with a central site and some neighbouring sites embedded in a bath, is chosen. There are several interaction parameters in the model.

2. The URG is used as a 'solver' for the impurity, to establish the flow diagram and the low energy spectral features.

3. By use of translation operators the results for the impurity are used to construct lattice properties.

If this is right, let me use this to frame questions for the Mott transition in the two dimensional half-filled Hubbard model which is the aim of chapters 4-5.

The 2D Hubbard model has parameters U and t, with the chemical potential set at $\mu = U/2$. There is only nearest neighbour hopping, so there is a nested Fermi surface in the band limit. Now the questions:

1. Is there an unique impurity model, in terms of geometry, that describes the lattice? The thesis uses a $1+4$ geometry, one can also imagine a $1+8$ or $1+12$ .. geometry.

2. For a particular choice of geometry there are more than two parameters. The geometry used in the thesis has an interaction J_K and an attraction W. Are these uniquely determined by the U and t?

3. The impurity model is much more complicated than the usual single impurity Anderson model (SIAM). The URG is used to solve it. Is that an exact RG or is some truncation involved? The original SIAM was solved 'exactly' through numerical RG by Krishnamurthy, Wilkins and Wilson. Does the URG work at the same level?

4. The RG focuses on low energy physics, while the spectral function has high energy features as well. Methods like DMFT - more about which later - capture both the low energy and the Hubbard band features. What about the present method?

5. The lattice 'phase diagram' at half filling is just a line, with U/t as axis. It goes from a Slater insulator at small U/t to the Heisenberg-Mott insulator at large U/t. There is no room for a metal, pseudogapped or otherwise, at $T = 0$. While the flow diagram of Fig.5.4 is fascinating, what are the corresponding Hubbard parameters?

These questions need not be answered in writing but would be taken up during the viva. Now we move to the chapter wise comments, which mainly expand on the issues above.

Chapter 1: The chapter does a good survey of Mott phenomenology. The thesis proposes a new auxiliary-model framework whose ambition extends beyond a new computational algorithm. Such a proposal requires a correspondingly detailed discussion (or at least tabular comparison) of existing approaches. In particular, the introduction contains relatively little critical discussion of the methodology and results of single-site DMFT, cluster extensions (CDMFT, DCA), variational cluster approaches, and other theoretical descriptions of Mottness.

Consequently, it remains difficult for the reader to identify precisely where the proposed auxiliary-model framework fits within the existing methodological landscape. Exactly which limitations of current impurity-based approaches motivate the present construction?

Chapter 2: This is the core chapter on methodology of the URG. The algebra is documented in great detail. However, for a reader the starting question would be what is the controlled approximation? Successful many-body methods are associated with a identifiable control parameter or asymptotic limit: for example, for the Wilson NRG it is the logarithmic energy-scale separation, for DMFT it is the locality of the self energy as $d \to \infty$, for tensor-network methods it is the low-entanglement structure, etc. What controls the URG for the extended impurity models? If there is an approximation, that needs to be clearly identified.

Chapter 3: Here the Mott MIT is solved on the infinite coordination Bethe lattice. This was also the first model solved within DMFT, three decades back. As a result one naturally expects a detailed comparison with the established DMFT solution. Discussion of the following quantities would significantly strengthen the chapter (i) quasiparticle weight, (ii) evolution of the spectral function, (iii) critical interaction strength, (iv) self-energy, (v) finite temperature behaviour (if feasible), and (vi) the computational complexity. Also, is there any conceptual difference that E-SIAM offers with respect to DMFT?

Chapter 4: Connecting impurity to lattice through translation operators. One qualitative question: while the $1 + 4$ cluster, for example, and the associated translation operators preserve the symmetry of the 2D lattice (modulo any long range order), does the process adopted generate the exact Greens functions of the lattice model? It certainly does not do so in methods like CDMFT, there is always loss of information.

Chapter 5: MIT on the 2D square lattice. This problem occupies a central position in the modern theory of strongly correlated electrons. The questions are below:

1. Benchmarks: The 2D Hubbard model has been investigated using a variety of high-accuracy numerical methods, including auxiliary-field quantum Monte Carlo, tensor-network approaches, exact diagonalisation, dynamical cluster approximation, cellular DMFT, variational cluster approximation, etc. Benchmark comparisons involving quantities such as the ground-state energy, double occupancy, quasiparticle weight, spectral function, self-energy, and spin correlation would have helped establish the validity of the auxiliary Hamiltonian description.

2. Uniqueness of the reconstruction: In principle, one could imagine alternative embedded clusters of different size or geometry while employing the same translation-operator formalism. For example, one might consider $1+4$, $1+8$, or larger auxiliary clusters. If the formalism defines a systematically controlled hierarchy, progressively larger clusters should converge towards identical lattice observables. Discussion of such a convergence hierarchy would be useful.

3. Interpretation of the Auxiliary Phase Diagram: The RG flows reveal a remarkably rich fixed-point structure. Viewed as the phase diagram of the auxiliary Hamiltonian, these results constitute an interesting contribution to quantum impurity physics. The interpretation as the phase diagram of the square-lattice Hubbard model is less straightforward. The microscopic Hubbard model is specified by the parameters (t, U), whereas the auxiliary Hamiltonian is described by the larger parameter set $(U_d, W, J_K, V, \ldots)$. Although the relation

$$U = U_d - W$$

connects the two descriptions, it does not uniquely determine the auxiliary Hamiltonian from the microscopic Hubbard interaction. The mapping from the Hubbard model to the auxiliary-cluster description appears non unique.

4. Role of the Auxiliary Couplings: The phase diagram is organised in terms of auxiliary couplings such as W and J. These quantities determine the infrared fixed-point structure and therefore the physical interpretation of the various phases. Are uniquely determined by the microscopic Hubbard Hamiltonian?

5. Relation to the half-filled Hubbard Model: The calculations are performed at half filling. For the nearest-neighbour square-lattice Hubbard model, long-range antiferromagnetic order is expected at zero temperature owing to perfect Fermi-surface nesting. The phase diagram presented here instead emphasises Fermi-liquid, pseudogap-metal and Mott-insulating fixed points. Has magnetic symmetry breaking been deliberately excluded? Similarly, the interpretation of the pseudogap phase requires clarification, since pseudogap behaviour is usually discussed in the context of doped cuprates rather than the half-filled nearest-neighbour Hubbard model.

Chapter 6: Modeling heavy fermions. This is done through a two orbital model, with interactions U_f and U_d, and auxiliary interactions J and W_f. The URG is performed on this, and translation operators are applied to obtain the lattice physics. The questions/comments:

1. Maybe one could have had simpler notation for the electronic operators, I struggled to figure out what $c_{Z,\sigma,f}$ might be.

2. The standard model for heavy fermions is the Anderson lattice. If eqn3.2 is the lattice model here, I guess the Anderson lattice corresponds to $U_d = 0$, $W = 0$ and $J = 0$. Yet the phase diagrams are plotted in terms of these parameters. How do we connect to Anderson lattice results?

3. There is detailed spectral work by D. E. Logan and coworkers on DMFT of the Anderson lattice using a local moment approximation. Do the spectral results here have any correspondence with them?

Chapter 6: Entanglement in 2D free fermions: I do not have any comments on this chapter.

As stated earlier, the thesis presents detailed original and careful work on a family on impurity problems using a novel RG technique. The work is documented in painstaking detail and deserves the award of a PhD degree. My questions are mainly on the connection of the impurity results to the standard correlated lattice models. These can be responded to in the viva.

Reviewer Type:	**International**
Reviewer Name:	**Prof. Subhendra D. Mahanti**
Reviewer Desination:	**Professor**
Reviewer Email:	**mahanti@pa.msu.edu**
Reviewer Phone No.:	**1-5178845633**
Reviewer Address:	**Department of Physics and Astronomy Michigan State University East Lansing, MI 48824-1116, USA Phone: +1-517-884-5633 (field below unable to accommodate)**
Reviewer Accepted/Rejected:	**Accepted**
Reviewer Selected For Viva voce:	**No**

Review of the thesis by Abhirup Mukherjee, IISER, Kolkata

Title: A new Auxiliary Model Approach to Fermionic Criticality: Insights in Mottness in Strongly Correlated Systems

Recommendation

The thesis is the result of very high-quality research on some of the challenging theoretical problems in condensed matter physics. The work is extensive and contains several new ideas. Although difficult to read, I found the work extremely interesting and strongly recommend its acceptance.

In the following I give my general comments with some questions which I would like Mr. Mukherjee to respond to during his oral presentation.

The thesis of Abhirup Mukherjee involves the development of a new auxiliary model method for studying translation invariant models of strongly-interacting electrons. In addition, it contains an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with emphasis on what happens at a critical Fermi surface. The primary focus is on understanding the underlying subtleties of Mott physics, how a strongly interacting fermi liquid changes to a phase with a pseudo gap (Mott metal) to finally a Mott insulator.

The procedure followed in the thesis is to first construct a quantum impurity model and then solve the model using an impurity solver. The author has used the unitary renormalization group (URG) method to solve the impurity problem at different stages. First, a dimension $D=\infty$ model where all the correlations are local and there is no information about the local geometry of the lattice in which the impurity is present. Then they embed the impurity in a 2D lattice (here they look at a square lattice). This gives him an understanding of the importance of non-local correlations which are present in a real interacting 2D many body system. After understanding the intricacies of the embedded impurity model, Abhirup looks at the real lattice problem by simply translating the solutions of the embedded impurity model to an infinite lattice. The idea is novel and interesting but the method of first solving a cluster problem and then translating its solutions to generate the infinite lattice problem <u>is an Ansatz</u>. But seems to be quite powerful in understanding Mott physics.

The thesis consists of 8 different chapters. The first two chapters, I and II are very well written, summarizing the history and current situation in the understanding the physics of strongly interacting systems, particularly the Mott physics. The suggestion that the “pseudo gap” phase seen experimentally is a distinct phase of strongly interacting system is both intriguing and reasonably convincing. The methods used in the thesis (URG method) have been extensively discussed in the past by the same group and are by now well founded. I am not going to make many comments on these two chapters except that I enjoyed reading the introduction very much and liked the summary of the developments in the field over last several decades.

Chapter I – Introduction

Chapter II – Methods and Preliminaries

Chapter III - Kondo Frustration via Charge Fluctuations: A Route to Mott Localization in Infinite Dimensions

In this chapter several interesting new ideas are discussed by introducing an extended single impurity Anderson model H_A, called $H_{E\text{-}A}$. The extension involves two extra terms. One is an explicit antiferromagnetic exchange coupling between the impurity spin and the spin of the bath at the impurity site, characterized by a parameter J. The other is a term like the U term of the Anderson model, but for the local charge density of the bath electrons, characterized by the parameter U_b. But unlike U which is positive and large, the magnitude of the parameter U_b is much smaller than U and it can be either positive (physically obvious) or negative (more subtle). This parameter, when negative, leads to Kondo frustration and helps in Mott localization. The purpose of this generalization of the SIAM, as the author points out, is to understand the dominant fluctuations at the heart of the Mott M-I transition and the physical processes that become important as one approach it.

The main results of the URG analysis in the limit of large U depend on the two quantities, the exchange coupling J and the parameter $r = |U_b|/J$.

$0 < r < r_{c1}$: the $J{-}V$ model; V (hybridization between impurity and bath) and J are both relevant, but U is irrelevant; spin and charge delocalization on the impurity; spin-charge mixing in ground-state. **Interacting Fermi liquid**.

$r_{c1} < r < r_{c2}$: the $J - U_b$ model; J is relevant, but V and U are both irrelevant; charge localization and spin delocalization on the impurity; singlet ground-state. **Insulating** (effect of large enough U_b) but spin mixing between the impurity and the bath.

$r_{c2} < r$: the $U - U_b$ model; U is relevant, but V and J are both irrelevant; spin and charge localization on the impurity; local moment ground-state. This phase describes a local **Mott insulator** on the impurity site, characterized by vanishing local double occupancy in ground state

From these calculations the author makes some general statements about the transition from a local metal to a local insulator, coexistence of local metallic and local insulating phases, and the possibility of a non-Fermi liquid behavior.

I found this chapter to be extremely well written with underlying mathematics and physics clearly explained. Although I didn't check the mathematical derivations, I assume they are correct because the author has published these results in peer reviewed journals.

What is the physical difference between "Insulator with spin mixing between impurity spin and bath spin" and "Mott Insulator". Explain

What is the physics of charge isospin Kondo coupling K between the impurity and the bath: needs to be explained better.

Chapter IV – A New Auxiliary Model Approach for Interacting Electrons: Mapping Impurity Models to Lattice Models

While this modified impurity model captures well the Mott metal-insulator transition of the Hub bard model in infinite dimensions, it lacks any knowledge of the underlying lattice geometry and is therefore not suited to address the phenomenology of more realistic low-dimensional models.

I have a problem with chapter 4 section 4.2.2, Periodization of Auxiliary Model into Extended Hubbard Model using the lattice translation operator on the impurity cluster model. The operator $c_{r_{d\sigma}}$ is a destruction operator for an impurity electron located at the lattice site r_d (originally site 0 of the impurity model). Also, in the impurity model there is a destruction operator associated with bath electrons associated with the same site, $c_{0\sigma}$. There are N such bath destruction operators for N lattice sites. The number of fermion destruction operators is therefore N+1. When one translates the impurity over the N lattice sites there are 2N fermion destruction operators, not N as given in equations 2.6 and 2.7. What am I missing. I would have expected a 2-band extended Hubbard model, one for the "impurity" and the other for the bath. (Later chapter discusses such a model).

All the discussions are associated with an extended 2D Hubbard model. Are you assuming that the solution of this Hamiltonian is simply the periodic translation of the solution of the impurity embedded in a 2D lattice. Is this an ansatz or are there some justifications for this approach?

Chapter V - Mott Criticality as the Confinement Transition of a Pseudo gap-Mott Metal: Insights on Mott Transition and Pseudo gap in Two Dimensions

Main results of the lattice embedded extended Anderson impurity problem and its lattice extension using translation Ansatz are discussed in detail in this chapter.

As pointed out earlier, the two important parameters of the extended Anderson impurity model are J and W. J is the exchange (antiferromagnetic) coupling between the impurity spin and the conduction electrons and W, which takes into account the correlation of the bath electrons which are in the neighborhood of the impurity. Unlike U which considers the correlation of the impurity electron and is positive and large, W is negative and small. Magnitudes of J and W are of the same order and they compete with each other. The origin of such a small attractive interaction between bath electrons when they are near the impurity is still not very clear to me. But as the URG analysis shows, this parameter plays a profound role in Mott physics.

I found Figure 5.4 to be very illuminating. It summarizes the essence of Mott physics. Two of the three fixed points, Fermi liquid and Mott insulating states, are well studied. The novel feature is the condition of sufficiently large attractive W. It is not clear to me how such an attractive interaction appears in a simple 2D Hubbard model. But the most interesting observation is the appearance of a third fixed point, pseudo gap fixed point in a finite range of W.

The other interesting feature of the calculations are the results shown in Fig. 5.6. Clearly the effect of lattice geometry in the momentum dependence of the coupling constants J(**k**,**k**') and W(**k**, **k**') shows in the spin-spin correlations and mutual information I_2.

Is the pseudo gap fixed point associated with insulating and spin mixing as discussed in the previous chapter. Can you describe the spin density distribution and spin-spin correlation function associated with this fixed point. What role does the lattice symmetry play in the nature of this pseudo gap fixed point. What will happen if the 2D lattice is triangular or hexagonal?

I had some problem with the strong conclusion "Thus, the drastic change in nature of the real-space excitations- from locally well-defined Landau quasiparticles of the FL to the increasingly nonlocal excitations of the NFL Fermi arcs in the PG phase-appears to be dictated by a topological principle and, therefore, robust under renormalization. This signals the NFL Fermi arcs of the PG as an emergent phase of strongly interacting quantum matter- which we dub the Mott metal- that is the parent metal of the MI."

What is the role of lattice dimensionality (D=2 vs 3) and lattice geometry (square vs triangular or Kagome') on this conclusion?

Chapter VI - Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localization

A two heavy fermion layer (f and d) Kondo impurity model has been studied to see how the Kondo physics gets modified as one increases the coupling between the two layers. Starting from a two-band Anderson model each band coupled to its own bath, they use a strong correlation limit (large U_f and U_d) to look at the two coupled Kondo layers.

Various properties of the models have been investigated by varying the bath correlation W_f (of the f-layer), assuming the bath correlation of the d-layer $W_d = 0$, and the inter-layer exchange coupling $J_{\perp}$. URG gives a phase diagram with several competing states. For large $J_{\perp}$, the state is always a Kondo insulator due to particle-hole symmetry, but the nature of the Kondo insulator changes depending on W_f. For small W_f, the Kondo insulator displays momentum-space anisotropy in terms of a selective opening of the hybridization gap at various parts of the Fermi surface. It is conjectured that the momentum-anisotropic entry into the Kondo insulator is likely to affect the small to large Fermi surface transition observed in the Kondo breakdown QCP class of heavy fermion systems away from half-filling.

Fig. 6.4 is quite illustrative. It provides an excellent picture of the effect of coupling between two Kondo systems on the behavior of the fixed-point values of the three Kondo coupling constants, Jd , Jf and $J\perp$. The $f-$layer Kondo coupling is highly suppressed by increasing both $|Wf|$ and $J\perp$. The layer with Wd=0 with coupling Jd is mildly suppressed by $J\perp$. The inter-layer coupling $J\perp$ itself grows as its bare value is increased. As one increases $J\perp$ heavy fermion transition goes from intra-orbital to inter-orbital Kondo screening. This is quite physical.

What happens for large W_f and one increases $J\perp$?

The author concludes this chapter with suggestions for further extension to his work. He thinks the appearance of long-range correlation at the transition from Fermi liquid to a different phase might be relevant to the observation of strange metallic behavior in heavy-fermion systems at high temperatures above the quantum critical point. But all the calculations are at T=0K.

I enjoyed going through this chapter very much. The two-band model has rich physics and has much wider application to many realistic systems. Not simply Kondo physics. I think this will encourage further research in this field.

Chapter VII - Holography of Entanglement in 2D Free Fermions: Emergent Geometry and Fermionic Criticality

I did not review this Chapter. Had some difficulty going through it. I have no comments on

Chapter VIII - Summary, Conclusions and Future Work

This chapter is very well written. Although I have not worked directly on the URG Easy to get the essential contributions of the extensive work of the author. for a researcher who has not worked URG method.

Reviewer Type: **Indian**
Reviewer Name: **Siddhartha Lal**
Reviewer Desination: **IISER, Kolkata**
Reviewer Email: **slal@iiserkol.ac.in**
Reviewer Phone No.:
Reviewer Address: **Indian Institute of Science Education And Research Kolkata,**
Mohanpur - 741246, Nadia
West Bengal, India
Reviewer Accepted/Rejected: **Accepted**
Reviewer Selected For Viva voce: **No**

भारतीय विज्ञान शिक्षा एवं अनुसंधान संस्थान कोलकाता, मोहनपुर-741246,
INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH KOLKATA, Mohanpur – 741246

EVALUATION REPORT OF THESIS

Name of the student:	Abhirup Mukherjee
Registration Number:	18IP014
Title of the Thesis:	A New Auxiliary Model Approach To Fermionic Criticality: Insights on Mottness in Strongly Correlated Systems
Name and Affiliation of the Reviewer:	Dr. Siddhartha Lal, Supervisor of the student

Recommendation (please tick the appropriate statement):

1. The thesis is recommended for the award of the degree without any correction. ☑
2. The thesis is recommended for the award of the degree provided the following points (as listed overleaf) are clarified during Viva-Voce.
3. The thesis is recommended for the award of the degree with minor revision (as listed overleaf/in the thesis)
4. Substantial revisions involving rewriting of one or more chapters are necessary (mentioned in detail overleaf) without, however, doing any further research work.
5. The thesis is not acceptable in the present form and needs to be rewritten. However, it reveals sufficient quality and quantity of work to warrant the student being given an opportunity for further research work and/or reinterpretation of results (see overleaf for details).
6. Not recommended.

Place: IISER Kolkata Campus, Mohanpur, West Bengal.

Date: 27th July, 2026

Siddhartha Lal

(Signature with Seal)

See overleaf

==

EVALUATION REPORT:

Abhirup Mukherjee's doctoral thesis reports research on building a framework towards the understanding of criticality and the various phases of quantum matter that can be attained in systems of interacting electrons at zero temperature. He has, at various points of his doctoral tenure, investigated the physics of Kondo screening and its breakdown in various quantum impurity models, the phenomenology of the Mott insulator-metal transition and the scaling of many-particle entanglement with its energy in a system of interacting electrons in two spatial dimensions as an explicit analytic manifestation of the holographic principle. Below, I lay out a chapter-wise assessment of each of these works.

Chapter 1 is an introduction to strongly correlated electron systems, focussing on the theme of the Mott metal-insulator transition central to the thesis. Considerable attention has been given to coverage of various models and methods that have been studied in the vast literature on the subject. This then motivates the questions that are dealt with in the thesis, as well as motivates the development of new methods for answering these questions. Chapter 2 lays out in details various methods that have been employed in the thesis (excluding the tiling method that has been developed by the author, and presented in Chapter 4). This includes the unitary renormalisation group (URG) method that was developed earlier in my research group; Abhirup's detailed exposition of these methods is welcome, as it makes the thesis largely self-contained.

The focus of Chapter 3 involves understanding the Mott insulator-metal transition of the half-filled Hubbard model on the Bethe lattice in infinite dimensions that is captured by dynamical mean-field theory (DMFT). For this, Abhirup has proposed a minimal effective impurity model that captures the phenomenology of this well-known Mott transition. This involves extending the standard Anderson impurity model Hamiltonian to include an explicit Kondo coupling, as well as a local on-site correlation on the conduction bath site connected directly to the impurity. For the case of attractive local bath correlations, a detailed URG analysis of the extended Anderson impurity model sheds new light on several aspects of the DMFT phase diagram. For example, the T=0 metal-to-insulator ground state quantum phase transition (QPT) is preceded by an excited state quantum phase transition (ESQPT) where the local moment eigenstates are emergent in the low-lying spectrum of the Fermi liquid. Long-ranged fluctuations are observed near both the QPT and ESQPT, suggesting that they are the origin of the quantum critical scaling observed recently at high temperatures in DMFT simulations. The T=0 gapless excitations at the QCP display particle-hole interconversion processes, and exhibit power-law behaviour in self-energies and two-particle correlations. These are signatures of non-Fermi liquid behaviour that emerge from the partial breakdown of the Kondo screening.

Abhirup then extends this line of investigation to a study of the Mott transition in an extended Hubbard model in two spatial dimensions at zero temperature. For this, he first lays out the development of a new auxiliary model approach in Chapter 4: he begins by fleshing out in detail the lattice embedded quantum impurity model that is appropriate for his study of the Mott MIT, and follows it up by formulating the tiling approach that is crucial for translating various quantities of the fixed-point impurity model (obtained from a URG analysis) onto that of the

See overleaf

full lattice extended Hubbard model. In Chapter 5, he has chosen to focus on understanding the origin of the pseudogap and strange metal phases of hole-doped Mott insulators by answering two key questions. First, is the pseudogap related to a precursor phenomenon in a symmetry-preserved Mott insulator? Second, is the strange metal a novel scale-invariant form of long-range entangled strongly interacting electrons? Abhirup has helped develop a new theoretical framework in dealing with these questions, and obtained rigorous answers that display their origin in a Mott transition of a particle-hole symmetric extended Hubbard model that includes nearest neighbour spin exchange interactions. His work has provided several exciting insights on the pseudogap phenomenology.

First among these is that the onset of a pseudogap in a strongly correlated Fermi liquid is signalled by a systematic momentum-dependent depletion of Kondo screening that suppresses quasiparticle weight and generates antinodal Luttinger surfaces. The pseudogap then emerges as a long-range-entangled "Mott metal'' with nodal arcs and holon-doublon non-Fermi-liquid excitations. The continuous confinement of holon-doublon pairs in passage through the pseudogap phase then realises Mott's vision of a repulsion-driven insulator. Remarkably, Abhirup finds that the exactly solvable (Hatsugai–Kohmoto) model describes the scale-invariant nodal non-Fermi liquid metal at the Mott critical point, displaying features that are universal throughout the pseudogap phase. In this way, his work demonstrates a unified symmetry principle that links Fermi and Luttinger surfaces, providing thereby a clear organisational framework for pseudogap phenomena, strange metals and Mott criticality. The continuous Mott transition uncovered in his work differs fundamentally from local quantum-critical scenarios inferred from other auxiliary-model methods such as DMFT and its cluster variants.

In Chapter 6, Abhirup follows up on this line of investigation by carrying out an auxiliary model analysis of a bilayer extended Hubbard model appropriate for the heavy fermion problem. Essentially, this is a two-orbital lattice problem in which one orbital is strongly correlated and on the verge of Mott localisation while the other is weakly correlated. The inter-orbital electronic hybridisation leads, in the limit of strong correlations in one of the orbitals, to the physics of Kondo screening and heavy fermions. By choosing both layers to be at half-filling, the strong Kondo screening regime corresponds to the formation of a Kondo insulator. An analysis of the problem is then carried out both in infinite dimensions, as well as in two spatial dimensions, revealing that the transition into the Kondo insulator involves passage through a Mott metallic phase (as observed in the Mott problem discussed in Chapter 5). Abhirup has presented several important preliminary results in this chapter, and is presumably completing this work for publication.

In Chapter 7, Abhirup has carried out an investigation of how the scaling of multipartite entanglement of a system of massive Dirac fermions in (2+1) dimensions under a sequence of renormalisation group transformations in momentum space can be considered as the generation of a holographic dimension. Geometric measures defined in this emergent space can be related to the RG beta function of the spectral gap, hence establishing a holographic connection between the spatial geometry of the emergent spatial dimension and the entanglement properties of the boundary quantum theory. Abhirup showed, analytically, that changing the boundedness of the holographic space involves a topological transition accompanied by a critical Fermi surface in the boundary theory.

See overleaf

Further, he demonstrated that this results in the formation of a quantum wormhole geometry that connects the UV and the IR of the emergent dimension. The additional conformal symmetry at the transition also supports a relation between the emergent metric and the stress-energy tensor.

Given the adiabatic continuity that links Fermi liquids to the non-interacting Fermi gas, it is expected that these results will be equally valid for the fixed-point effective theory for the Fermi liquid. This also suggests that the results for the wormhole geometry and the stress-energy tensor relation at the critical Fermi surface apply very generally to phase transitions that destabilise and gap out the Fermi surface. Specifically, this has profound implications for metal–insulator transitions driven by strong correlations. Abhirup has also observed the existence of a hierarchical sequence of quantities related to the multipartite entanglement defined at a particular RG step that are topologically non-trivial, all of whom encode the same Luttinger volume for the gapless systems that have been studied. This displays that the topological stability of a gapless system of interacting electrons is encoded within the structure of its entanglement.

Finally, in Chapter 8, Abhirup has summarised the questions addressed in the thesis as well as the main answers obtained from his work. Additionally, he has also offered a few open directions for further work. In summary, Abhirup Mukherjee's doctoral research provides the foundations of a theoretical framework for the systematic analysis of the quantum criticality and emergent phenomena in systems of strongly interacting fermions, and especially those involving metal insulator transitions arising from Mott localisation. For this, Abhirup has adopted a variety of analytical as well as numerical approaches including the development of a new auxiliary model method. The thesis offers several novel insights and results on many paradigmatic problems of strongly correlated fermionic quantum matter, and has led to the publication of several papers. Furthermore, the approach taken here is promising for the further investigations of various open questions in other strongly interacting systems of electrons. I recommend that Abhirup Mukherjee's dissertation be accepted for the awarding of the Ph.D. degree without any corrections.

Siddhartha Lal

27th July 2026.

See overleaf